Variation Models

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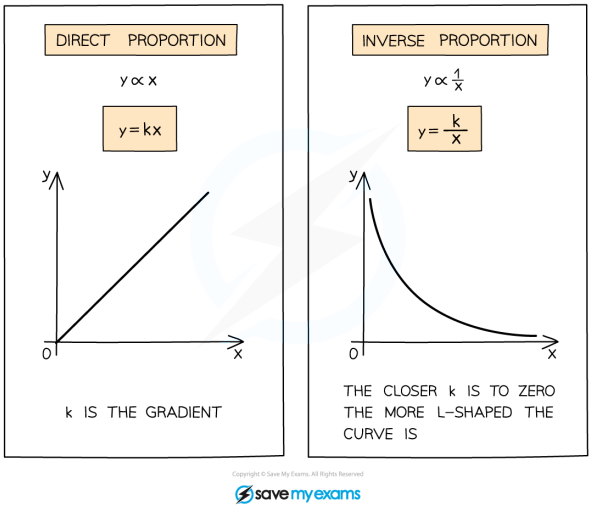
Variation Models

What is a variation model?

- A **variation**, or **proportion**, model describes how one quantity varies with another
- For example:
  - $y = kx$ is a **linear** variation model and shows **direct proportion**
  - $y = kx^2$ is a **quadratic** variation model
  - $y = kx^3$ is a **cubic** variation model
  - $y = \frac{k}{x}$ is a **reciprocal** variation model and shows **inverse proportion**



How do I use a variation model to find missing data values?

- The equation of a variation model is an equation of a curve or line that models a set of **data**
  - Substitute in the x-values to find the y-values (or the other way around)

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

substitute a known pair of x and y values into $y = kx$ to find k
    - You can then use the full equation to find any missing values

Worked Example

Consider the following table of data.

x	1	3	q
y	36	p	9

Given that y is inversely proportional to x, find the values of p and q.

As y is inversely proportional to x we can write the equation

$$y = \frac{k}{x}$$

Substitute the pair of known values into the equation to find k

$$36 = \frac{k}{1}$$
$$k = 36$$

So the full equation of the variation model is

$$y = \frac{36}{x}$$

Substitute in $x = 3$ to find p

$$p = \frac{36}{3} = 12$$

$$\boldsymbol{p = 12}$$

Substitute in $y = 9$ to find q

$$9 = \frac{36}{q}$$
$$9q = 36$$
$$q = \frac{36}{9} = 4$$

$$\boldsymbol{q = 4}$$

How do I identify the best variation model for given data?

- You may be asked to select the **most suitable variation model** for given data
- Using the data to **sketch a rough graph** is one way to do this
    - You could do this on your graphic display calculator

choose the most suitable from several) you should **test the values** using the given data

- **Substitute** the value of x into the suggested model, and see how close the result is to the actual value of y in the table
- Do this for at least 3 data points
- You will then be able to judge if the suggested model is appropriate, or select the best one

Worked Example

Consider the data in the table below.
Decide which out of a linear, quadratic, or cubic variation model would be most suitable for the data, and use this to find an equation for y in terms of x.

x	2	4	6
y	20	80	180

Start by checking if the model is linear, $y = kx$

Using the first data point to find k

$$20 = k \times 2$$
$$k = 10$$

Use this to check the second data point

$$y = 10x$$
$$y = 10(4) = 40$$

Not close to 80, so not linear

Consider if the model is quadratic, $y = kx^2$

Use the first data point to find k

$$20 = k \times 2^2$$
$$20 = 4k$$
$$k = 5$$

Use this to check the second data point

$$y = 5x^2$$
$$y = 5(4)^2 = 80$$

It works for the second point, now check the third

$$y = 5(6)^2 = 180$$

Also correct for the third point, so this is an appropriate variation model

A quadratic variation model is suitable

Its full equation is $y = 5x^2$

If a cubic model is tested, $y = kx^3$, the first point produces the equation $y = \frac{5}{2}x^3$ but this does not model the other two

Worked Example

Two variation models are suggested for the data in the table below.

$$\text{Model A: } y = \frac{400}{x^2}$$

$$\text{Model B: } y = \frac{800}{x^3}$$

x	5	6	10	15
y	25	15	3	2

Determine which variation model is more suitable.

Add a row to the table for each of the suggested models

Use the same given values of x (5, 6, 10, 15) to calculate the y values that each model predicts

E.g. To find the y value that model A predicts for $x = 5$

$$y = \frac{400}{5^2} = 16$$

x	5	6	10	15
y	25	15	3	2
Model A: $y = \frac{400}{x^2}$	16	11.111...	4	1.777...
Model B: $y = \frac{800}{x^3}$	6.4	3.703...	0.8	0.237...

Compare the modelled y values to the actual y values

The values for Model A are much closer to the actual values than Model B

Model A is more suitable

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